Decision log: the embodied-fly path and its branch points

*Every crossroads on the way from the chip-to-animal study to the embodied fly, with the options that were open, the one taken, why, and where to branch from to explore another. Results are not restated here: each stage's numbers live in `report/ROADMAP_EMBODIED_FLY.md` §7, cited to `outputs/metrics.json`, and each stage's registration in `preregistration/`. Commits are on master unless marked otherwise. Decisions taken by the user are marked **(user)**; the rest were taken by the agent and are open to revision.*

How to explore another branch

Check out the commit named in the decision, create a branch, register the alternative in `preregistration/` before running it (fresh seeds if the original result has been seen), and run it with its own command or `python -m neuromps.fly.pipeline --only <step>`.

Decisions

D1. Which fly connectome release the model runs on

- **Options:** FlyWire 630 (the published model's release), FlyWire 783 (the current one).
- **Taken:** 783 for the studies, with a registered 630-vs-783 replication of the published sugar result. **Why:** the current release, with the published release kept as a check.
- **Revisit at:** `fly_chip_to_animal` registration; `neuromps.fly.connectome.load_630`.

D2. Which body simulator for the embodied fly

- **Options:** FlyGym / NeuroMechFly v2 (needs Python 3.12), the FlyMimic musculoskeletal leg loaded in plain MuJoCo, flybody.
- **Taken:** FlyMimic in plain MuJoCo (pinned `mujoco` as the optional `embodied` extra); flybody later for the proboscis. **Why:** muscle-driven, runs in the project's Python 3.11, no new framework. **Revisit:** a FlyGym environment would give six muscle-driven legs only if its musculoskeletal model covers them.

D3. After stage 1 failed: what to change first

- **Options:** rescale the male-CNS synaptic weight further; drive the male nerve cord from descending neurons with the brain removed; switch connectome.
- **Taken:** the nerve cord from descending neurons (stage 1c), then the FlyWire brain joined to it through type-matched descending neurons (stages 1e, 1f). **Why:** the male brain avalanches under the published parameters (`fly_spread`); each connectome is used where its model is validated.

D4. Stage 1e's precondition failed on an absolute rate line

- **Taken:** stage 1f, a relative precondition and depth of reach, on fresh seeds; 1e stays uninterpretable. **Why:** the absolute line had no basis.

D5. Stage 2a failed: how to look for a walking rhythm (user: sweep)

- **Options offered:** fix adaptation parameters from literature; sweep them and report the whole sweep. **Taken (user):** the sweep (stage 2c), with a registered selection rule and confirmation.
- **Revisit:** published adaptation values, if any appear for fly leg or premotor neurons.

D6. The glutamate sign rule (user: go ahead)

- **Taken by the agent:** two bounding rules (glutamate excitatory onto motor neurons only; and everywhere), each with the adaptation grid (stage 2d). **Unexplored:** under the corrected rhythm measure the selection rule would pick (glutamate everywhere, 16 mV, 100 ms); it was not confirmed on fresh seeds.

D7. A different neuron model (user: look at how others make flies walk)

- **Found:** the walking flies online use trained controllers (Eon Systems, flybody); only Pugliese et al. (2025) get a rhythm from nerve-cord wiring, with a firing-rate model.
- **Taken:** reimplement their model from the paper and configuration (their code is unlicensed).
- **Size stand-in (stage 2e):** total synapse count, because the male CNS has no volumes. This starved DNg100; stage 2f drove every command neuron to the same rate; stage 2g used their front-leg network and recruitment rule; stage 2h a geometric input search.
- **Unexplored:** neuron volumes (a future male-CNS release, or MANC volumes through the male-CNS-to-MANC matching); their own network files under a licence.

D8. A flaw in the registered rhythm measure

- **Taken:** a registered audit with a peak-requiring measure (prominence 0.1, calibrated on synthetic signals only), re-scoring stages 1d-2e on their original seeds; registered outcomes kept as recorded, the documents carry both. Stage 1d's D2 withdrawn.
- **Revisit:** prominence 0.05 or 0.2 give the same calibration results; a different peak criterion could be registered.

D9. Where to close the loop first (stage 3)

- **Options:** the front leg (muscle-driven, thinly traced proprioceptors), a middle or hind leg (well traced, but only position-servoed in flybody), the proboscis.
- **Taken:** the front leg through the claw neurons; the unknown flexion/extension tuning bracketed by two assignments and a sham; then a perturbation test (stage 3b).
- **Unexplored:** a hind-leg loop through flybody's servos and its 22 claw neurons; hook and club neurons; campaniform sensilla.

D10. Drugs in the loop (stage 4)

- **Taken:** each knob acts on every network of a trial; movement disruption uses

log2((x + 0.01 mm)/(x_vehicle + 0.01 mm))	with the parent's 0.5 line, 0.05 mm as sensitivity.

- **Disclosed issue:** at full GABA-A block the reach depth is negative, so that condition's index averages the two proboscis terms. **Revisit:** a registered rule for negative depths.

D11. Checking against real flies (stage 5)

- **Taken:** a search protocol registered before any search; every candidate classified.
- **Unexplored:** neuron-specific simulations of the identified-neuron studies (DSOG1, 13A/13B) instead of class-wide knobs; a satiety state for the feeding model.

D12. Release (stage 6)

- **Taken:** a pipeline that reruns every later fly study as its own process, wired into `run_all.py` as a full-run-only `embodied` stage. **Why:** one command reproduces every number; separate processes keep the connectomes from accumulating in memory.

D13. Working overnight without the user (user: continue, commit, push, update docs)

- **Taken:** "push" read as merging finished work into master through a pull request, because GitHub keeps master only; every crossroads tonight is added here.

D14. The precision a chip needs, at the movement level (overnight)

- **Options:** leave stage 4a's timing result as it stands; compute, as the parent study did at the neural level, the chip's change at the dose where movement first fails.
- **Taken:** the latter, as an exploratory analysis (both inputs had been seen), added to the pipeline as `stage4a_precision`. **Why:** the paper's assay-design message rests on the precision a chip needs at behavioural onset; this extends it to movement at no simulation cost.

D15. Confirming the cell the corrected measure selects (overnight)

- **Options:** leave stage 2d's alternative selection unconfirmed (as the audit registered); confirm it on fresh seeds.
- **Taken:** confirm it (registration `embodied_stage2d_confirm_corrected.json`, commit 8495b57), with the corrected measure throughout. **Why:** it is the one loose end the audit named, and it is cheap. **Not taken tonight:** a hind-leg closed loop through flybody's servos (D9), a volume-based size rule (D7), neuron-specific drug simulations (D11), each larger.

D16. Bringing the embodied fly into the technical report and a paper draft (overnight)

- **Taken:** Act VIII of `report/TECHNICAL_REPORT.md` (§27-§31, every number cited, the uncited-number budget unchanged) and `report/PAPER_DRAFT.md` (abstract to methods, every number cited). **Why:** the user asked for a paper and for the report to keep being retouched; the draft is a starting point for the team, not a submission.

D17. Testing the model against identified-neuron experiments (overnight)

- **Options:** a hind-leg closed loop (D9), a volume size rule (D7), simulating the identified neurons that stage 5's published studies manipulated (D11).
- **Taken:** the identified neurons (stage 5b, registration commit c20fa08): the left front leg's GABAergic 13A and 13B hemilineage neurons, silenced (outputs removed) or activated (150 Hz), in the grooming relay of stage 1f. **Why:** it turns stage 5's indirect comparison into a direct one with data already in hand, and it is affordable overnight. **Not possible:** Pool et al.'s DSOG1 neurons are not annotated in FlyWire. **Detail:** 73 of the 76 annotated 13B neurons of that neuromere are in the nerve-cord network used.
- **Revisit:** subsets of 13A/13B (the study manipulated sub-populations), pulsed activation (the study's 70 ms pulses induced grooming-like rhythms), the femur-tibia angle as a readout ("legs locked in flexion").

D18. Reading the 13A/13B manipulations at the joint (overnight)

- **Options:** pulsed 13A activation (the study's grooming-inducing protocol; a leaky integrate-and-fire network has no rebound, so it would very likely fail by construction); a hind-leg closed loop (larger); the joint readout the study actually used.
- **Taken:** the joint readout (stage 5c, registration commit 04a2387, fresh seeds): femur-tibia angle under the same arms as stage 5b. **Why:** stage 5b's mismatch could be a readout mismatch (reach versus joint posture); this tests that directly and cheaply.
- **Process note:** a test edit for this step did not apply and a piped test run hid it; fixed in commit fd51761. Test runs before commits are no longer piped.

D19. Closing the loop on a well-traced leg (overnight)

- **Options:** stop stage 3 at the front-leg limit; close the loop on the right hind leg (25 claw neurons) through flybody's servoed tibia; build a muscle-driven hind leg (no such model at hand).
- **Taken:** the servoed hind leg (stage 3c, registration commit 22b082e): the walking command holds the right hind tibia extended (its extensor pool is tonically active, stage 2a), a 20 ms torque pushes it, and the return is timed under the resistance and assistance assignments and a sham, as in stage 3b. **Why:** it tests whether the front leg's failure was the thin sensor tracing, with the one body that has the leg. **Cost:** feedback changes a commanded angle, not muscle force; a servo's own stiffness sets much of the return.
- **Revisit:** a muscle-driven hind leg (if FlyMimic or another model adds one); the walking command's other legs.

D20. Testing stage 3c's exploratory difference on its own (overnight)

- **Options:** report stage 3c's resistance-against-sham difference as exploratory and stop; test it alone on fresh seeds.
- **Taken:** test it (stage 3d, registration commit 8ea194a): resistance against sham, paired by seed, 32 fresh seeds, a one-sided sign test and a 5 ms margin. **Why:** the difference was the one sign that the push-following part of the feedback acts on a well-traced leg, and it was seen in a noisy, unstable regime, so only a registered test on new seeds can say whether it is real.
- **Outcome:** 0 of 2 (stage 3 stays a documented negative). **Revisit:** a muscle-driven hind leg.

D21. After stage 3: replicate the headline before anything new (overnight)

- **Options:** new science that needs a model not at hand (a muscle-driven hind leg, neuron volumes for the size rule); 13A/13B subsets or pulsed activation (a leaky integrate-and-fire network has no rebound); paper and slide writing only; a registered replication of stage 4a, the paper's headline, which ran on one set of 8 seeds.
- **Taken:** the replication (stage 4b, registration commit b23f6a3): stage 4a's movements on 16 fresh seeds, against the chip detections of the parent design's own fresh-seed replication, with each movement left out in turn. **Why:** it is the result a reviewer will press first, it is cheap (about 15 minutes), and the leave-one-out check names the assumption most likely to make it wrong (that one movement carries the disruption index).
- **Outcome:** 3 of 4 (registration commit b23f6a3). The chip is late in every family on the fresh seeds; without the bitter-gated proboscis GABA-A potentiation ties the chip.
- **Revisit:** a different impairment line for movement (the 0.05 mm sensitivity is reported); a finer dose grid for GABA-A potentiation and excitability block, where the margin is one step.

D22. Resolving stage 4b's one-step margins (overnight)

- **Options:** leave the two thin margins reported; a finer dose grid inside them, as the motor-level tie-break did; a different movement impairment line (a new rule chosen after the results, so not taken).
- **Taken:** the finer grid (stage 4c, registration commit f387845): GABA-A potentiation at gaba_scale 1.6-1.9 and excitability block at threshold_scale 1.24-1.36 and 1.44-1.56, with stage 4b's movement seeds and both of the parent's chip seed sets, the coarse levels rerun as a reproduction check. **Why:** stage 4b showed that in two families the headline rests on a single step of the dose grid, and a finer grid is the direct way to ask what happens inside it.

- **Outcome:** 3 of 4 (registration commit f387845). Excitability block resolves chip later against both chip runs; GABA-A potentiation stays chip later with the full index, and tied without the bitter-gated proboscis.
- **Revisit:** a fourth movement (for example a walking readout, once one exists) would tell whether GABA-A potentiation's lateness is specific to feeding.

D23. Stopping new science for the night (overnight)

- **Options:** further stages (each remaining branch needs a model not at hand or is expected to fail by construction); documents for the team.
- **Taken:** a slide outline for the deck (`report/SLIDES_OUTLINE.md`, every number cited), the README's document table brought up to date, and a morning summary in `PROGRESS.md`. **Why:** the headline has now been tested as far as the current bodies allow (stages 4a-4c), and the next scientific choices (D7 neuron volumes, a muscle-driven hind leg, a walking readout for drugs) are the team's to make.

D24. The volume size rule (user)

- **Options (D7's unexplored branch):** neuron volumes from a future release, or MANC volumes through the male-CNS-to-MANC matching; found instead: the male CNS v1.0 neuPrint neuron table (same bucket, same CC-BY release) has a voxel count per body.
- **Taken:** fetch that table whole (4.65 GB, source `malecns_sizes`), checked against the bucket's CRC32C, because the bucket publishes no MD5 for it **(user: full file, not a column-only range read)**; rerun stage 2h with the published rule (stage 2i, registration commit d03ee27) on fresh draws. A numpy CRC32C was written rather than adding a dependency.
- **Outcome:** 6 of 7 (registration commit d03ee27): the rhythm needs the circuit and promotor and remotor keep a reliable, non-antiphase offset; flexor-extensor alternation fails.
- **Revisit:** normalizing by the median over the whole nerve cord rather than over the network (stated choice: over the network, as for the synapse-count stand-in).

D25. Checking that stage 2i comes from the neurons' own volumes

- **Options:** report stage 2i as it stands; name the assumption most likely to make it wrong and test it (the project rule for results that look good).
- **Taken:** stage 2j (registration commit 1d1d644), fresh draws: the volume rule replicated, volumes shuffled across neurons (a fresh permutation per draw), and normalization by the nerve cord's median instead of the network's. **Why:** if shuffled volumes do as well, the result is about the spread of sizes, not the identified neurons; the normalization was a stated choice the published description does not fix.
- **Outcome:** 2 of 4 (registration commit 1d1d644). Shuffled volumes give no rhythm; the cord-median normalization changes nothing; the rhythm and the circuit's necessity replicate, but the promotor-remotor offset does not reach the registered reliability, so it is not claimed.
- **Revisit:** a registered test of the offset alone with more draws, if it matters for the paper.

D26. Walking as a drug readout (user)

- **Options (from D22's revisit):** a fourth movement for the drug design; the only walking readout at hand is stage 2i's fictive front-leg rhythm (rate model, volume sizes, no body).
- **Taken:** stage 4d (registration commit 73ffde6), fresh draws: the drug knobs applied to the walking network (weights scaled by presynaptic class, as in the spiking model; the excitability knob scales the activation threshold), each draw's DNg100 input found under vehicle and held fixed, onsets on stage 4c's combined dose series. **Why:** it asks whether GABA-A potentiation's lateness is specific to feeding and whether the headline holds for a rhythm.

- **Stated choice:** the threshold knob maps to the rate model's theta; fixed inputs, so a drug changes the response, not the stimulus.
- **Outcome:** 2 of 4 (registration commit 73ffde6). GABA-A and GluCl block break walking first; acetylcholine block is chip-first; GABA-A potentiation and the threshold knob leave walking intact; with walking as a fourth movement the chip is still late in every family.
- **Revisit:** a readout that respects the bimodal draws (the fraction of rhythmic draws) registered as its own test; a threshold mapping that also shifts the drive.

D27. Retesting the acetylcholine exception with a readout that respects the draws (user)

- **Options:** keep stage 4d's median readout; a fraction-of-rhythmic-draws readout; a per-draw paired analysis.
- **Taken:** the fraction readout on fresh draws (stage 4e, registration commit 180476f). **Why:** stage 4d's draws are bimodal, the median sits between the modes, and the exception entered the abstract; the fraction is the plainest readout of a bimodal outcome and uses the rhythm line registered since stage 2g.
- **Outcome:** 2 of 4 (registration commit 180476f). GABA-A and GluCl block break walking first; the acetylcholine exception does not replicate: mild block raises the share of rhythmic draws, and the two-sided index counts that as a change before the chip. The exception is withdrawn from the abstract, discussion, README and slides.
- **Revisit:** a one-sided walking index (only loss of rhythm counts) would be a new rule chosen after seeing this result; register it only with a reason that does not depend on the outcome.

D28. The paper and new research move to a private repository (user)

- **Options:** keep everything public; make the repository private (would break the competition's links and the hosted demo, and erase stars and watchers); rewrite public history (destructive, and GitHub keeps every pull request's commits regardless); keep new work private from here on.
- **Taken:** new work private from here on. The public repository `Vodolay/AI4S` keeps the hackathon deliverables and everything already published (up to pull request 26). A private repository `Vodolay/AI4S-paper` mirrors public master and adds a `paper-draft` branch holding the paper draft, its outline, the slide outline (D23) and any new research toward the paper; those three documents were removed from public master. **Why:** the user wants the preprint and the research leading to it kept private until submission; what was already public stays so, and its history timestamps the work.
- **Revisit:** when the paper is submitted, `paper-draft` merges into public master.

D29. Recovering the ToxCast components that have no level-0 rows

(Numbered D55 in the project's private research log, where it was first taken.)

- **Found:** the invitrodb v4.3 dump's level-0 table lacks 8 of the 17 acute MEA components, among them firing rate, spike number and burst number; level 3 still has their rows.
- **Options:** use only the 9 components with level 0 (no firing rate); ask the EPA for the missing rows; recover the wells from the dump's id allocation.
- **Taken:** recovery (`neuromps.etl.toxcast_recover`). Level-0 ids come in contiguous runs of equal length per component and batch, so a component's run start is predicted from the reference component's; rval is recovered from level 3 exactly (rval = bval + resp x (pval - bval) / 100). Wells are assigned by position in the reference run; for all 7 missing electrophysiology components concentrations agree on every row and each date's stored baseline is reproduced from the assigned wells on every date. Four present components have their runs in another order (both checks fail on them), so present components keep their own level-0 wells.

- **Assumption recorded:** positions within a date are not checked by the baseline, so the physical-plate assignment of a recovered well rests on the run order alone.
- **Revisit at:** `data/interim/toxcast_mea/recovery_checks.json`.

D30. Testing the twin's dose response against real concentration-response wells

- **Found:** §33 showed the twin's knobs get the direction of the EPA acute screen's three positive controls right, but each is run at one saturating concentration, so §34 recorded the open question: nothing there tests the twin's dose response. The screen also runs about 500 compounds over seven concentrations each, with a viability readout on the same plates.
- **Options:** leave the dose response untested (the deck's closing slide already lists it as a next step); match knob levels to concentrations by potency (needs an IC50 per compound and per platform, which §14/HD8 showed does not transfer); match by rank and score only what a rank match can carry.
- **Taken:** the rank match, registered in `preregistration/twin_toxcast_dose_response.json` before any treated-well value was computed, with the compound list and one class label per compound fixed in the registration from published primary-target annotation. Scores are a Spearman rank correlation of the twin's matched curve against the real one per compound and readout, which readout moves first, and two nulls the twin must beat: the compound's own real firing-rate curve (the §33 follow-firing-rate rule, extended to curves) and the twin's curve under a knob the compound is not labelled with. **Why:** the knob scales and the concentration axis share no unit, and every registered score is invariant to a monotone map between them, so a rank match supplies exactly what the scores can use.
- **Registered before looking:** firing rate counts as already seen (a private analysis looked at it) and is excluded from the shape tests; bins are retained only up to the first that the screen's own viability readout flags, so that a collapse at the top concentration cannot be read as pharmacology; a compound needs four retained bins to be scored.
- **Assumption recorded:** the rank match assumes the screen's concentration range covers a comparable span of the mechanism to the knob grid. Checked by rescoring with the grid's two strongest levels dropped (`sensitivity_truncated_grid`).
- **Revisit at:** `outputs/metrics.json` key `twin_toxcast_dose_response`, report §35.

D31. Recalibrating the twin's inverse model: more simulations, then an ensemble

- **Found:** the inverse model (one masked autoregressive flow, 2,700 training simulations, `twin_hd14_inference`) fails the rank-uniformity check for `gabaa_scale` and `inter_in_degree`, so the slide withholds both.
- **Options:** keep withholding them; correct the check for five parameters (Bonferroni would pass both, which lowers the bar rather than fixes the model); retrain the same flow on more simulations; average several flows; change the flow family.
- **Taken:** a registered test of the two remedies that do not touch the bar: about 15,000 fresh simulations (5x; truncated at 05:00 EDT on 2026-10-28 by a prefix rule fixed in advance) for the same flow, and an equal-weight ensemble of five flows on the same pool, both judged on 1,000 fresh held-out simulations by the original per-parameter test at 0.05, plus coverage, contraction against the old model on the same simulations, and a check near the real recordings. The flow family is not changed, so budget and averaging are the only differences. **Why:** a slide that withholds two of five parameters should say whether that is a limit of the budget, of the estimator, or of what the summaries carry.
- **Registered:** `preregistration/twin_inference_recalibration.json` (seven predictions and the assumption check), committed before any new simulation.

- **Outcome:** 2 of 7 (registration commit 0bc29aa, amended before any simulation in 176b36e). 51 of 60 training chunks finished before the cut. On fresh simulations the round-1 model fails all five parameters, not two; the ensemble keeps its sharpness and passes four of five on the 300-run test, but fails `inter_in_degree`, stays over-confident by coverage for four parameters, and takes about 2 s per posterior. Budget alone fixes neither failing parameter. Under the registered rule the slide quotes `threshold_scale` only. Report §16, change-log entry 45.
- **Revisit at:** `neuromps.twin.recalibration`; `twin_hd14b_recalibration` in `outputs/metrics.json`. Next branches, each to be registered with fresh seeds: a different flow family (neural spline flow) on the same pool, which would separate estimator from summaries; or summaries that carry connectivity more directly (the failure that neither remedy moves).